

C++ Lab Exercises

Week 15

1. Create a class called `time` that has separate `int` member data for hours, minutes, and seconds. One constructor should initialize the data to 0, and another should initialize it to fixed values. Another member function should display it, in 11:59:59 format. The final member function should add two objects of type `time` passed as arguments. A `main()` program should create two initialized `time` objects (`const`) and one that isn't initialized. Then it should add the two initialized values together using an overloaded `+` operator, leaving the result in the third `time` variable. Finally it should display the value of this third `time` variable. Make appropriate member functions `const`.
2. Modify the `FeetInches` class (in Sample Programs) so it overloads the following operators:
`<=`
`>=`
`!=`
Demonstrate the class's capabilities in a simple program.
3. Add a copy constructor to the `FeetInches` class. This constructor should accept a `FeetInches` object as an argument. The constructor should assign to the `feet` attribute the value in the argument's `feet` attribute, and assign to the `inches` attribute the value in the argument's `inches` attribute. As a result, the new object will be a copy of the argument object. Next, add a `multiply` member function to the `FeetInches` class. The `multiply` function should accept a `FeetInches` object as an argument. The argument object's `feet` and `inches` attributes will be multiplied by the calling object's `feet` and `inches` attributes, and a `FeetInches` object containing the result will be returned.
4. Make a `LandTract` class that is composed of two `FeetInches` objects, one for the tract's length and one for the width. The class should have a member function that returns the tract's area. Demonstrate the class in a program that asks the user to enter the dimensions for two tracts of land. The program should display the area of each tract of land and indicate whether the tracts are of equal size.